

Question ID: 76f29fa5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

The cost to rent a commercial fishing boat from a certain company is **\$950** for the first **2** hours and an additional **\$50** per hour for each hour after the first **2** hours. If the total cost to rent the commercial fishing boat from the company for t hours, where $t > 2$, is **\$1,100**, which equation represents this situation?

Answer

- A. $950(t - 2) + 50t = 1,100$
- B. $950(2t) + 50t = 1,100$
- C. $950 + 50(t - 2) = 1,100$
- D. $950 + 50(2t) = 1,100$

Correct Answer: C

Rationale

Choice C is correct. It’s given that the cost to rent a commercial fishing boat is **\$950** for the first **2** hours and an additional **\$50** per hour for each hour after the first **2** hours. It’s also given that t represents the total number of hours and $t > 2$. Therefore, the number of additional hours after the first **2** hours can be represented with the expression $t - 2$. The cost for these additional hours is **\$50** per hour, so the cost for the additional hours can be represented by the expression $50(t - 2)$. The total cost can be calculated by adding the cost for the first **2** hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 2)$. It’s also given that the total cost to rent the commercial fishing boat from the company for t hours is **\$1,100**. Thus, the equation that represents this situation is $950 + 50(t - 2) = 1,100$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.